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| **Fiscal Year:** | FY 2009 | **Task Last Updated:** | FY 09/02/2009 |
| **PI Name:** | Sutherland, Betsy M Ph.D. | | |
| **Project Title:** | DNA Damage Clusters in Human Cell Transformation Induced by Single or Multiple Space Radiation Ion Exposures | | |
| **Division Name:** | Human Research | | |
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| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
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| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | bms@bnl.gov | **Fax:** | FY 631/344-3407 |
| **PI Organization Type:** | GOVERNMENT | **Phone:** | 631/344-3380 |
| **Organization Name:** | Brookhaven National Laboratory | | |
| **PI Address 1:** | Dept of Biology, Building 463 | | |
| **PI Address 2:** | 50 Bell Ave | | |
| **PI Web Page:** | | | |
| **City:** | Upton | **State:** | NY |
| **Zip Code:** | 11973-5000 | **Congressional District:** | 1 |
| **Comments:** | PI Deceased as of October 2009. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2006 Radiation Biology NNJ06ZSA001N |
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| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Cucinott1a, Francis | **Contact Phone:** | 281-483-0968 |
| **Contact Email:** | noaccess@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | NOTE: PI Betsy Sutherland passed away in October 2009. Stefan Tafrov took over the project in December 2009. SEE TAFROV for reports from FY2010 forward. | | |
| **COI Name (Institution):** | | | |
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| | The space radiation environment poses long and short-term risks to human travelers. During a 3-year Mars mission, the nucleus of each cell in an astronaut's body will be hit by ~400 protons and 0.4 high energy, high atomic number (HZE) ions, plus high levels of protons during Solar Particle Events (SPEs). Predicting and mitigating cancer risk from complex radiation fields in outer space is essential for manned exploration of Mars, as well as long-term habitation of the moon. This requires understanding the roles and importance of damages that induce critical changes, cellular repair responses and potentially-oncogenic cellular alterations. Thus, understanding the biological consequences of multiple particle exposures is essential for assessing radiation risks of space travel. Our research focuses on understanding the consequences to human cells of exposure to the levels and kinds of radiations present on the moon and in deep space. We use a human primary cell neoplastic transformation model, in which irradiated human cells acquire growth | | |

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| **Task Description:** | properties of human cancer cells. We found that exposure to a moderate proton dose followed by a HZE particle gave higher transformation levels than the sum of the two exposures, indicating non-linear effects of multiple radiation exposures. We are continuing our studies of exposing human cells to multiple space ions including protons at low doses, in SPE-simulating proton fields and in combination with HZE ions. Cell survival and transformation levels are measured.<br>Space radiation induces complex DNA damages affecting both DNA strands/double strand breaks (DSBs) and clusters of damaged bases. We are measuring the induction, repair and levels of unrepaired clusters and DSBs, in order to correlate these data to cell killing and transformation. We developed methods for quantifying DNA bistranded clusters in human cells irradiated with low doses of low LET radiation. We showed that Fe ions, X- and X-rays induce clusters in cells with a linear dose-response and that clusters are repaired via repair-intermediate DSBs and abasic clusters. We recently developed a method for measuring DNA cluster levels in cells exposed to relevant occupational levels of low LET radiation, (mSv doses). However, the induction and repair of clusters in cells exposed to simulated complex radiation fields, and the effects of such radiation on induction of potentially oncogenic damage are unknown. We are initiating studies to determine whether levels of slowly-repaired or unrepaired, persistent clusters correlate with the human cell neoplastic transformation frequencies observed with mixed field radiations. We are determining whether conditions or agents that reduce the levels or types of endogenous or radiation-induced clusters also reduce transformation levels, thus suggesting approaches to mitigation of cancer risk during space travel. Melatonin, a human hormone, effectively scavenges radiation-induced hydroxyl radicals that damage DNA. Our preliminary data indicate that melatonin effectively reduces cell transformation by HZE ions. We propose to test its effectiveness in reducing biological damage induced in human cells by single or multiple space particles. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Research Impact: We are addressing Research & Technology Question 28a, (the effect of radiation quality and dose rate on carcinogenesis probability), using four hypotheses directly addressed by our four specific aims. These are: Hypothesis 1: Low proton fluencies will not induce cell killing, but may affect cell cycle progression and thus could affect transformation by later HZE particle hits. Specific Aim 1: Determine the effects, including transformation, induced by realistic space radiation exposures of protons (1, 2, or 4 protons per cell nucleus).<br><br>Hypothesis 2: Protons of the energy and LET (linear energy transfer) ranges present in SPEs will have similar effects on cell survival, transformation and DNA damage spectrum as high energy (1 GeV/n) protons. Specific Aim 2: Measure transformation, cell killing and DNA damage spectra and repairability of complex damages produced by protons of different energies in SPEs. Hypothesis 3: SPE beams will affect cell killing and transformation as the sum of the constituent photons at the same dose rate. Specific Aim 3: Compare biological effects of SPE beams with the sums of protons of similar energies and dose rates (and to 1 GeV/n protons at acute dose rates).<br><br>Hypothesis 4: Mixed field irradiations of 1, 2 or 4 H/nucleus followed by one HZE hit/nucleus will produce survival curves dominated by HZE effect; however, low H fluxes may induce repair and thus decrease transformation frequencies. Specific Aim 4: Determine the transforming and killing potential of dual hits by 1, 2 or 4 protons/nucleus subsequently followed by one HZE particle/nucleus.<br><br>Earth Benefits: This project is determining the effects of charged particles on human cells, specifically the killing properties of the radiation and its ability to induce cell transformation, a model for human cancer. Data obtained in this grant year show that both the killing ability and the transforming ability of the radiation increase with the amount of energy deposited by the specific type of particle, known as the radiation's linear energy transfer characteristic. These studies are directly applicable to life on earth, as charged particles are beginning to be used in tumor radiotherapy in humans. The advantage of particles over gamma rays or X-rays is better targeting of tumors and less damage to adjacent non-tumor tissue. Protons for cancer therapy are now available at several US medical centers and additional proton radiotherapy facilities are under construction or being planned. In addition, heavier particles such as carbon ions are in use in Europe and in Japan for radiation therapy, and offer the possibility of even better spatial targeting of tumors than is possible with protons. Concern has arisen concerning additional radiation generated by activation of material in the treatment room by the particles during their use for radiation therapy. Similar activation occurs in the target room during irradiation of experimental samples; this offers an excellent opportunity to characterize this additional radiation and to devise approaches to minimize it.<br><br>Innovative Technology For investigating differences in repair of X-ray (photon) vs. charged particle-induced DNA damages, we designed and developed a new dual-fluorescence tag system for visualizing and quantifying DNA repair. This assay replaces the use of radioactive DNA labeling with fluorescent labels, thus greatly decreasing preparation frequency and thus experimental costs as well as avoiding the use of radioactivity and eliminating disposal costs. This system won a Department of Energy Best in Class Pollution Prevention Award. Results from the use of this new technology to probe repair of space radiation-induced DNA damage are now published in Nucleic Acids Research.<br><br>We developed methods for quantifying DNA bistranded clusters in human cells irradiated with low doses of low LET radiation. We showed that Fe ions, X- and X-rays induce clusters in cells with a linear dose-response and that clusters are repaired via repair-intermediate DSBs and abasic clusters. We have recently designed and constructed a molecular reporter system to study the repair of complex DNA lesions. We have alsovrecently developed a method for measuring cluster levels in cells exposed to occupational levels of low LET radiation at milli-Sievert (mSv) doses. This will enable us to further characterize complex DNA damages formed by exposure to simulated space radiation fields. |
| | During this year of project, we are continuing to investigate five aspects of human cell transformation by space radiation and the ability of cells to repair space radiation damage: 1) effects of low proton fluences on human cell neoplastic transformation, 2) mechanisms of increased transformation by proton plus heavy charged particle irradiation, 3) efficiencies of different charged particles in inducing cell transformation, 4) effectiveness of simulated Solar Particle Event radiation in killing or transforming human cells, and 5) induction and repair of complex DNA damages in human cells.<br>To evaluate the oncogenic potential at the cellular level of space radiation fields, we are using a model system that incorporates several key elements of human exposure to radiation during space travel. First, the target cells are human primary cells (rather than rodent cells or immortalized cell lines), and the measure of radiation damage is a human |

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| **Task Progress:** | cancer surrogate, the acquisition of anchorage-independent growth by the cells. The ability to grow in a non-permissive, semi-solid medium is also termed “neoplastic transformation”. Second, we are examining three space-simulating regimes, a mixed field regime in which the cells are exposed first exposure to protons, then to a HZE particle; second, very low proton fluences, simulating the low proton dose rate found routinely in space; and third, a long-duration exposure to low energy protons, simulating the a SPE event. We are also testing the efficacy of different ions and energies in inducing soft agar growth in our cell system. To better understand the contributions of specific HZE particles to human cell transformation, we are currently continuing to determine the effectiveness of 1 GeV/n Oxygen Ions at Brookhaven National Laboratory's NSRL as begun in Summer 08 and plan to examine others as the ions become available at NSRL. We are also planning to extend our transformation system to other cells types, that are capable of becoming tumorigenic in people.<br><br>Effects of Low Proton Fluences on Human Cell Neoplastic Transformation<br><br>During routine space radiation exposures, many cells in a traveler’s body will be exposed protons but only occasionally to a HZE particle. Thus understanding the effects of dual hit H + HZE particle requires knowledge of the 1 H ion + 1 HZE ion hit per cell. Initial experiments on effects of single proton hits per cell were apparently confounded, by the activation of objects in the NSRL target room (including our sample holder) and by earlier proton exposure during the experimental day, with huge scatter and lack of reproducibility. We solved this problem by devising a sample holder that introduced a minimum of material into the beam: a gallows-shaped polyethylene support (that remains outside the beam). The samples are suspended in a zip-lock bag suspended from the gallows by plastic clips. We also introduced the use of opaque bags, as visible light impinging on cell culture medium can induce mutations in cultured cells. These changes, plus, improved irradiation procedures—tuning NSRL to protons the previous day, using a minimal tune in the morning, having first morning beam time, and carrying out the low fluence experiments at the beginning of our experiments allowed us to obtain monotonically-increasing responses to low fluence proton exposure.<br><br>Mechanisms of Increased Transformation by Proton plus Heavy Charged Particle Irradiation<br><br>We have previously observed that the timing between exposure of human cells to protons and to a single heavy charged particle (dual hit or mixed-field irradiation was) was critical --if this time window was less than one hour, the transformation efficiency was increased by about 3-fold. We have previously eliminated two hypotheses: alterations in the cell cycle, or changes in cell survival as mechanisms for the increased transformation levels we observe in our system. We are currently analyzing levels of 53BP1 focus formation in dually irradiated cells in order to measure cellular levels of repair resistance. Preliminary data suggests that fewer foci result from the dual beam exposures than either ion alone.<br><br>Ion Properties Affecting Human Cell Dual Irradiation Responses<br><br>Understanding the effects of space radiation on human cells requires that we understand the efficiency of different heavy charged particles in inducing cell transformation. This is essential for evaluation the mixed field irradiations that simulate space radiation. We found that the particles that deposit more energy (per particle) are more effective in inducing cell transformation. The relation between energy deposition and transformation efficiency is approximately linear. We have tested both iron and silicon ions in dual proton HZE particle experiments. We plan investigate whether other proton-HZE particle dual irradiations also result in increased transformation as other dual ion beams become available at NSRL.<br><br>Compared to the differences in energy deposition per particle for heavy ions, protons over a wide range of energies have energy deposition values that are much more similar. Prior to embarking on full SPE simulations, we tested the relative transforming efficiency of protons of decreasing energy. We observe that lower energy protons which deposit more energy as they traverse the cell have higher transformation efficiencies.<br><br>Effectiveness of Simulated Solar Particle Event Radiation in Killing or Transforming Human Cells<br><br>We are progressing with experiments simulating solar particle event (SPE) exposures to human cells. We tested monoenergetic protons in the range of 50 or 100 MeV/n H, predominant energies and biological effect among SPE particles. In initial experiments we examined both protons of both of these energies at dose rates approximating 1.65 and 33.65cGy/min for their relative effectiveness in cell killing and transformation.<br><br>Induction and Repair of Complex DNA Damages in Human Cells<br><br>An important question in evaluating space radiation risk is whether the biological effects of high energy protons (e.g., 1 GeV/n) are similar to those of ionizing photons (X and gamma-rays) so that the extensive literature on X- and gamma-rays can be used to evaluate human risk. A priori, it seems logical that ionizing radiation species of similar LETs would have similar biological effects. However, this approach does not consider effects of factors such as track structure. Since both the levels and composition of complex DNA damages (both double strand breaks and clustered damages) have long been considered to depend on track structure, it seemed possible that any differences in the effects of protons and photons might be reflected in the spectrum of DNA damages.<br><br>To test this possibility, we measured the levels of DSBs, Abasic clusters, and Oxidized Base clusters in DNA irradiated with one of a number of particles (LETs varying from ~ 0.2 to 150 keV/um) or X- or gamma-rays. These data will provide a test at the molecular level to predict the similarities or differences between photons and protons in producing biological damage, including cell killing and transformation. We plan to study other ions in this system as they become available at the NSRL.<br><br>NOTE: PI Betsy Sutherland passed away in October 2009. Stefan Tafrov took over the project in December 2009. SEE TAFROV for reports from FY2010 forward. |
| **Bibliography Type:** | Description: (Last Updated: 04/12/2011) |
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